

Deploying Web App on ECS - Step 7: Create ECS Fargate Service

To Begin with the Lab

Summary of the Lab

In this step, an **Amazon ECS Fargate Service** was created to run the verified task definition continuously. The service was configured with two desired tasks so the application would stay available even if one task failed. It was attached to the private subnets, reused the existing **ECS task security group**, and disabled public IP assignment because the application was meant to be exposed through the **Application Load Balancer** instead. The existing ALB, listener, and target group were selected so traffic could be routed to the ECS tasks automatically. After the service became active, both tasks were running and the application was confirmed by opening the ALB DNS name and testing an image upload to **Amazon S3**.

Understanding Amazon ECS Service with a Load Balancer

An **Amazon ECS Service** is what keeps container tasks running all the time instead of starting them once and leaving them alone. It exists so ECS can restart failed tasks, keep the desired number of tasks alive, and work with a load balancer for steady traffic distribution. In this lab, the service was connected to the **Application Load Balancer**, which lets users reach the app through the ALB DNS name while the tasks stay inside private subnets. The benefit is a reliable, high-availability setup where the app stays online and ECS handles task replacement automatically.

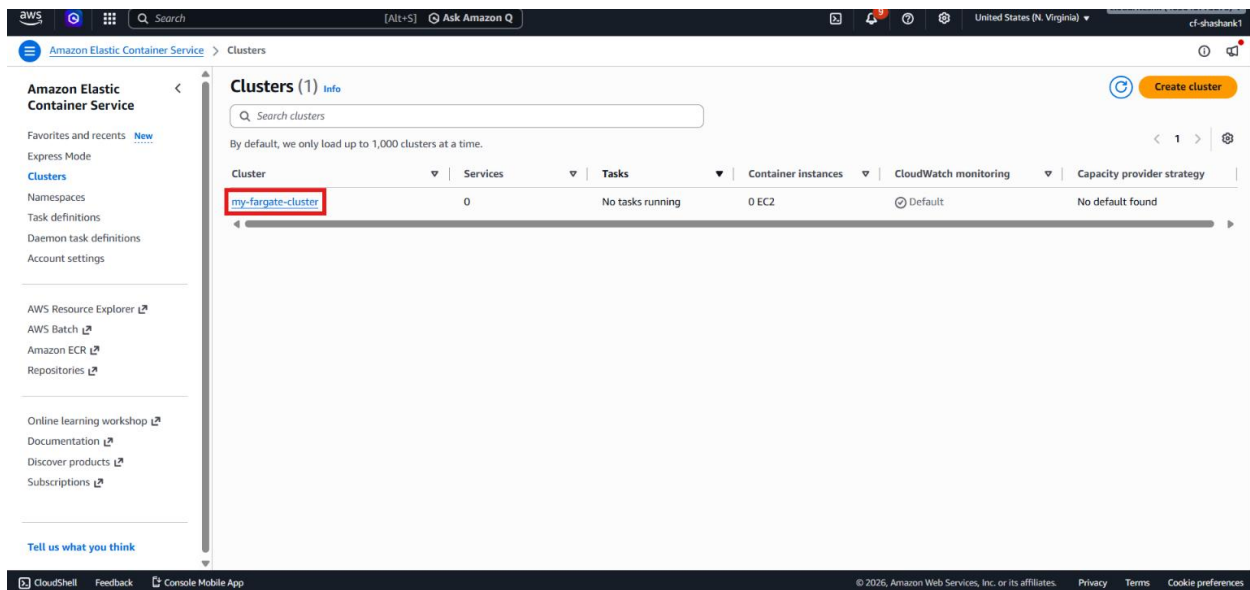
Example:

A PHP web app runs with two ECS tasks in private subnets. A user opens the application through the ALB DNS name, and the load balancer forwards the request to one of the healthy tasks. If one task stops, ECS launches a replacement and the ALB continues serving traffic through the remaining healthy task.

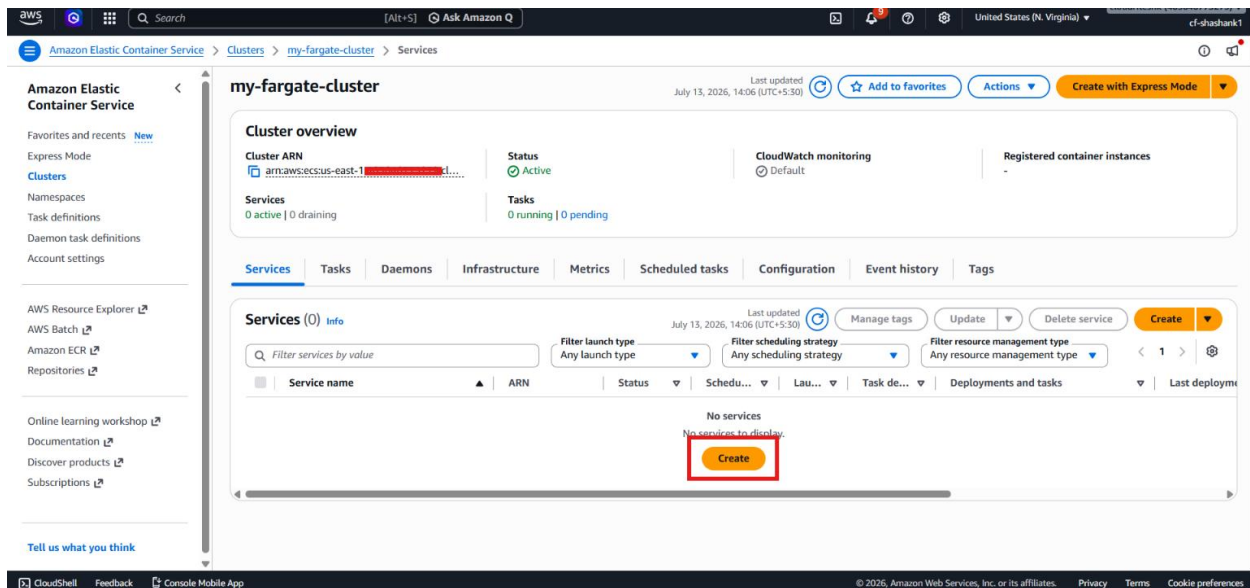
Lab Steps

Step 1 – Open the ECS service creation page

- Sign in to the **AWS Management Console**.
- Open **Amazon ECS**.
- Select the existing cluster.

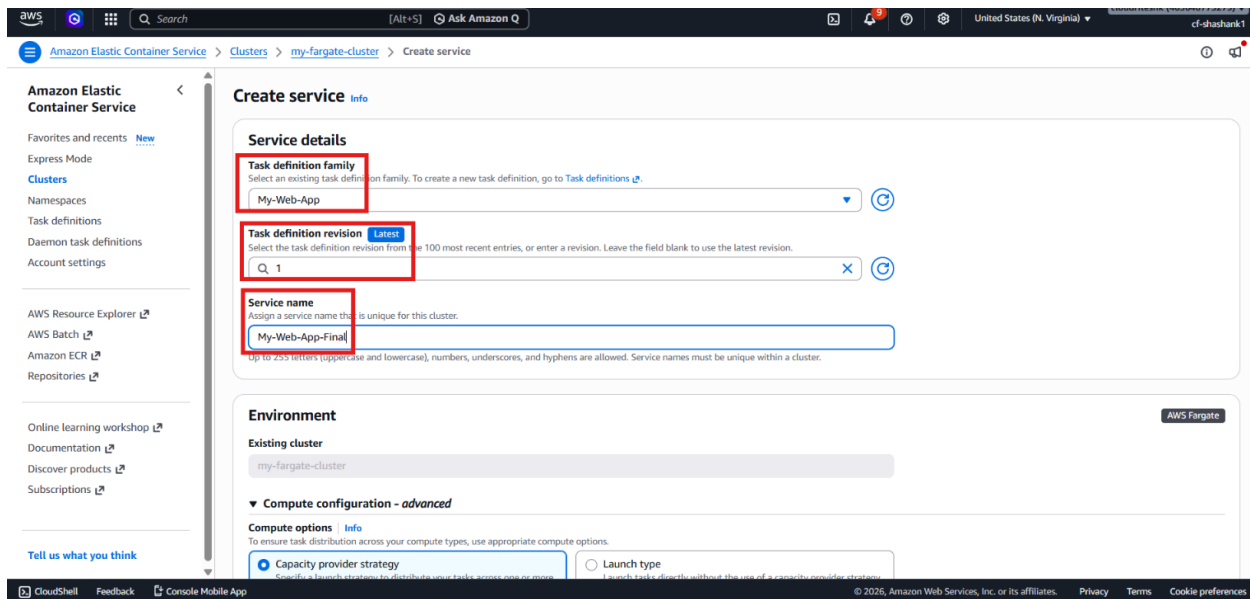


- Click **Create service**.

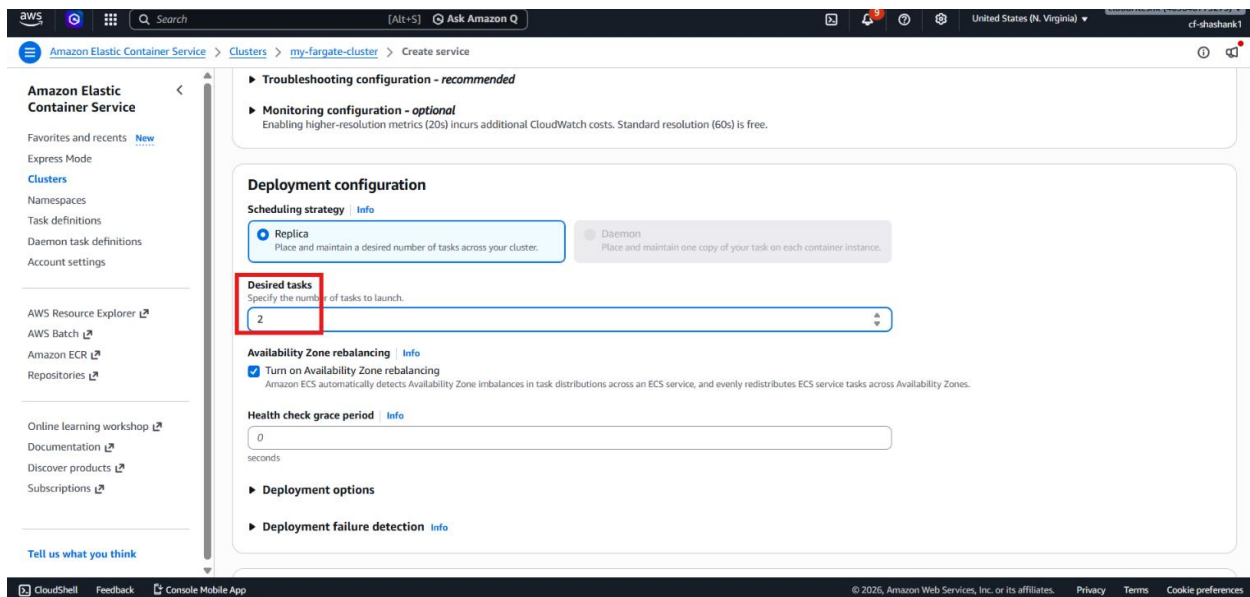


Step 2 – Select the task definition

- Choose the verified task definition family **My-Web-App**.
- Select the latest revision.
- Set Service Name as **My-Web-App-Final**.

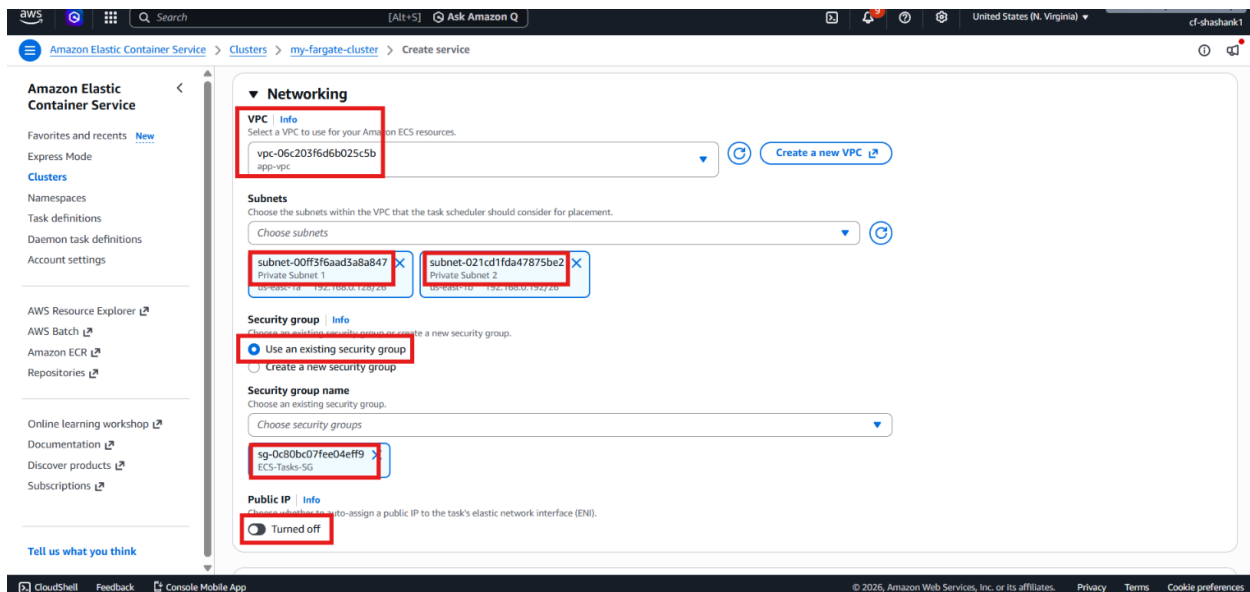


- Set the desired task count to 2.



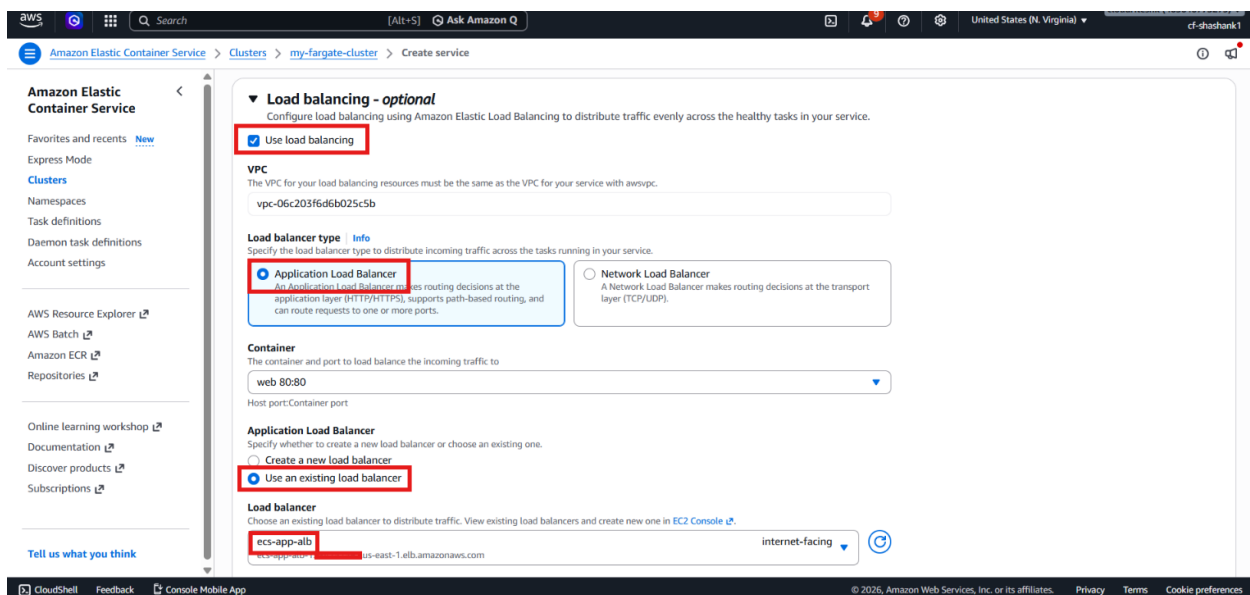
Step 3 – Configure the networking

- Choose the VPC we have created earlier.
- Select only the private subnets for the ECS tasks.
- Remove the public subnets from the service configuration.
- Select the existing **ECS-Tasks-SG** security group.
- Do not assign a public IP to the tasks.

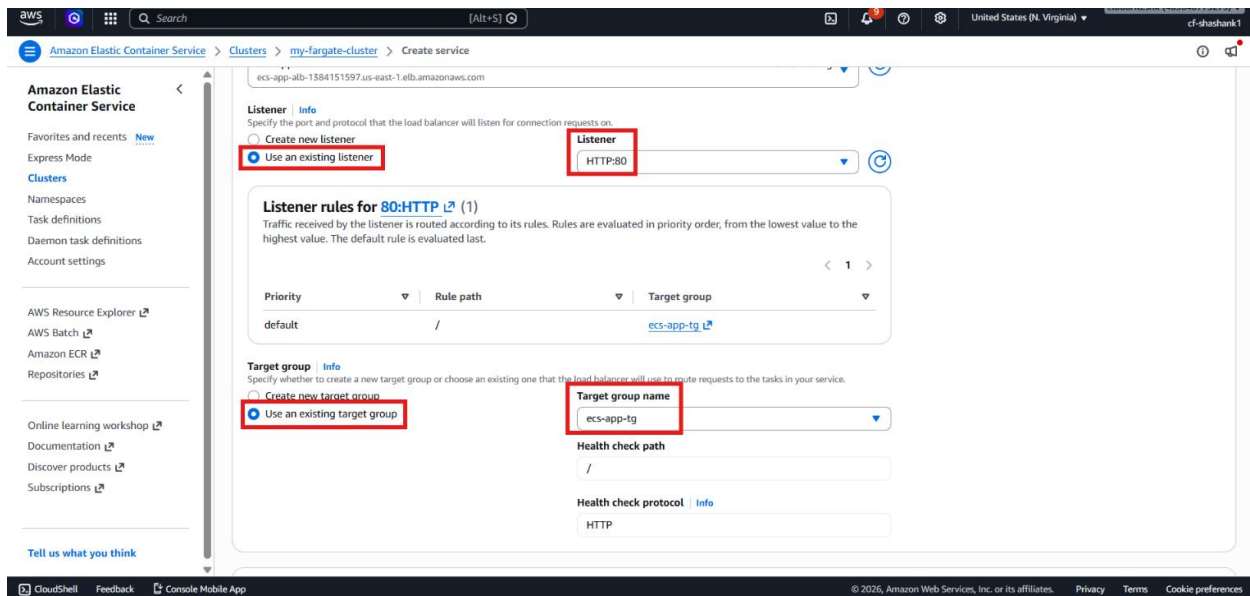


Step 4 – Attach the load balancer

- Enable Load balancing.
- Select **Application Load Balancer**.
- Choose **Use existing load balancer**.
- Select the ALB created earlier.

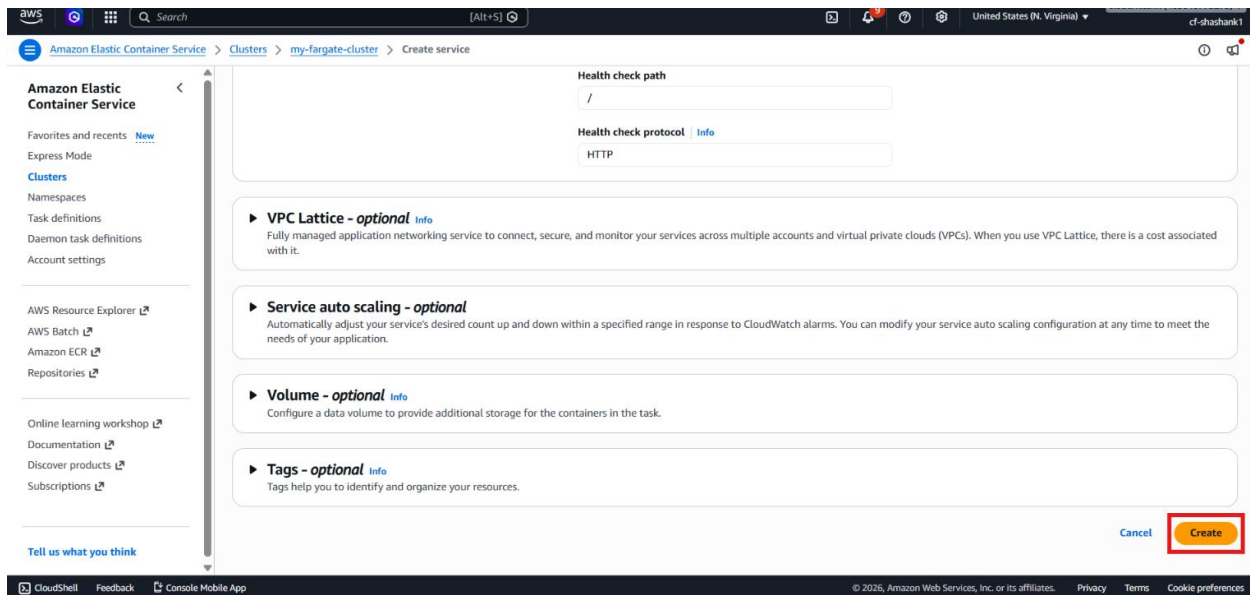


- Choose **Use existing listener**.
- Choose **Use existing target group**.
- Confirm that traffic on port 80 will be forwarded to the ECS tasks.



Step 5 – Create the ECS service

- Keep the deployment type as rolling deployment.
- Review the configuration.
- Click **Create**.

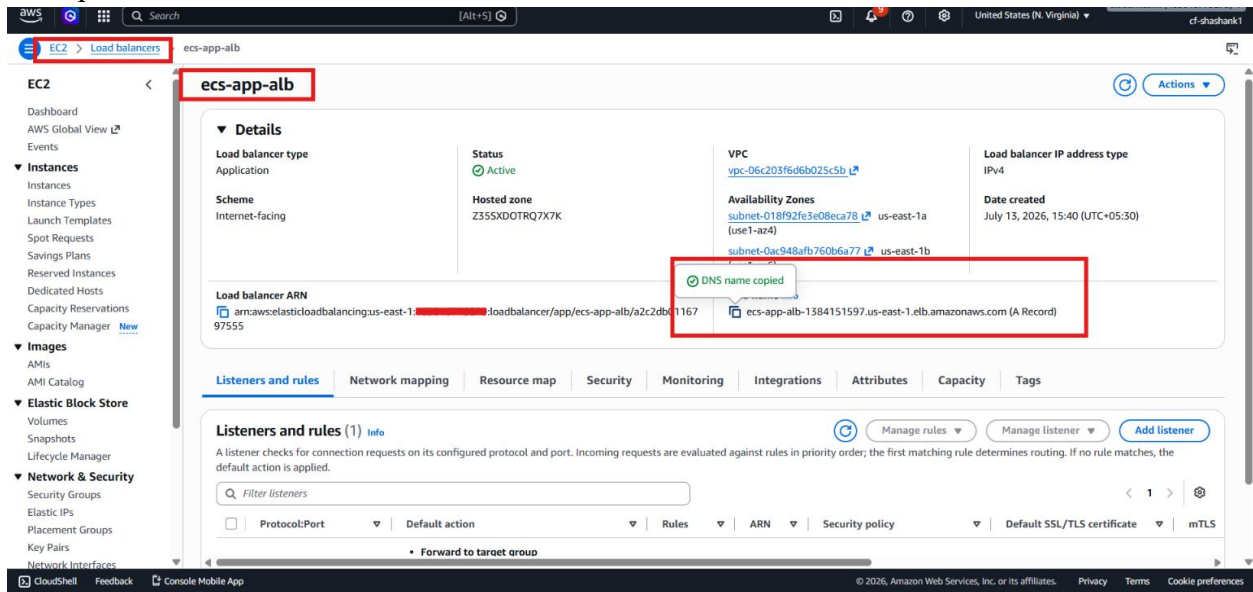


- Wait for the service to become active and the tasks to start running.

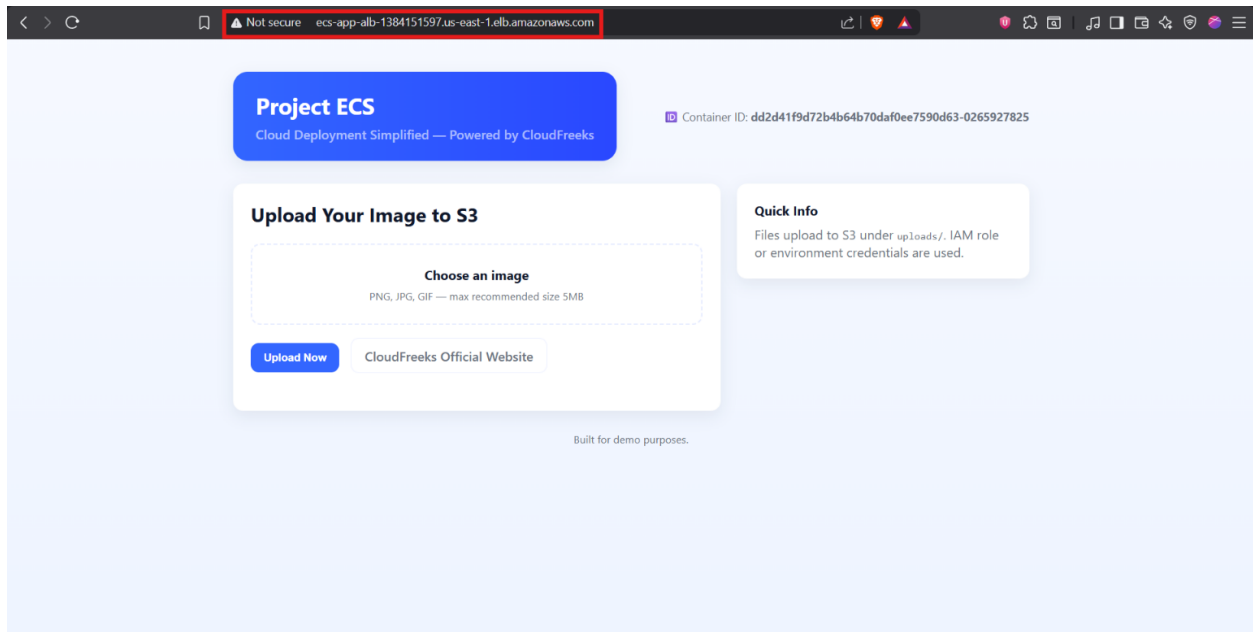
Step 6 – Verify the running application

- Wait until both desired tasks are running.
- Copy the ALB DNS name.

- Open the ALB URL in a browser.



- Confirm that the PHP application loads successfully.



- Upload a sample file and confirm that the file is stored in **Amazon S3**.

Step 7 – Verify target registration

- Open **EC2**.
- Go to the target group used by the ALB.
- Confirm that both ECS tasks are registered as healthy targets.

The screenshot displays the AWS Management Console interface for configuring a Target Group. The breadcrumb navigation shows the path: EC2 > Target groups > ecs-app-tg. The left-hand navigation pane includes sections for EC2, Images, Elastic Block Store, and Network & Security. The main content area shows the configuration for the 'ecs-app-tg' target group, including its IP address type (IPv4), protocol (HTTP), and port (80). It also shows the associated load balancer (ecs-app-alb) and VPC (vpc-06c203f6d6b025c5b). A summary section indicates 2 total targets, all of which are healthy. Below this, a table titled 'Registered targets (2)' provides details for each target, including IP address, port, zone, and health status. Both targets are listed as healthy.

IP address	Port	Zone	Health status	Health status details	Administrative override	Overri...	Anomaly detectio...
192.168.0.183	80	us-east-1a (...)	Healthy	-	No override	No overri...	Normal
192.168.0.196	80	us-east-1b (...)	Healthy	-	No override	No overri...	Normal

- Verify that the load balancer is routing traffic correctly.

Verification

- Verified that the ECS service was created successfully.
- Verified that the service used the correct ECS task definition revision.
- Verified that the service was configured with two desired tasks.
- Verified that the tasks were placed in private subnets.
- Verified that the **ECS task SG** was attached.
- Verified that no public IP was assigned to the tasks.
- Verified that the existing **Application Load Balancer** was attached.
- Verified that the target group registered both tasks as healthy.
- Verified that the application was accessible through the ALB DNS name.
- Verified that the file upload worked and the object appeared in **Amazon S3**.